

The Word-Complexity Ladder for Collatz Divergence: a Machine-Verified Rung Zero and a Mahler-Value Reduction of the Automatic Rung

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Status note (August 2026). The rung-one question left open here — the irrationality of the 2-adic Mahler value $F_2(8/9)$ — has since been settled and machine-verified: no orbit has the itinerary $\sigma^\infty(1)$ or any shift of it (`lean/Collatz/Rung1.lean`), and the truncation exponent 0.946 discussed in Section 6 is a barrier for truncation only; a degree-one Padé form suffices. See the companion paper *Rung One of the Word-Complexity Ladder* (`paper/rung1.pdf`). Rung zero, the reduction, and the calibration below stand as written.

Abstract

Order potential divergent orbits of the Terras map ($T(n) = n/2$ for even n , $(3n+1)/2$ for odd n) by the complexity of their parity itineraries. We prove the floor of this ladder formally: a divergent orbit's itinerary is never eventually periodic (machine-verified in Lean 4, axiom-free). We then reduce the next rung to a single arithmetical statement. For the fixed point w of the substitution $1 \mapsto 110$, $0 \mapsto 011$ — an aperiodic automatic word of parity density $2/3$, inside the supercritical band where divergent orbits must live — the generating function $F(z) = \sum w_i z^i$ satisfies the Mahler functional equation $F(z) = (z + z^2)/(1 - z^3) + (1 - z^2)F(z^3)$, and the unique 2-adic integer whose itinerary is w equals $-10 + \frac{5}{9}F_2(8/9)$, where F_2 denotes 2-adic evaluation. Consequently, if the 2-adic Mahler value $F_2(8/9)$ is irrational, no integer has itinerary w . The archimedean value $F(8/9) \in \mathbb{R}$ is transcendental by Mahler's classical method; the 2-adic value is the open case, and we prove that it is genuinely beyond naive methods: the truncation approximations to $F_2(8/9)$ have irrationality exponent exactly $(3 \log 2)/(2 \log 3) = 0.9464 < 1$, so no Liouville-style argument can decide the question — and the deficiency constant is, once again, the critical density of the $3x+1$ problem. The same reduction applies uniformly to every constant-length substitution with constant block weight in the supercritical band, where double (real and 2-adic) convergence of the relevant series is automatic.

1 Introduction

The Collatz conjecture asserts that iterating the Terras map

$$T(n) = \begin{cases} n/2 & n \text{ even,} \\ (3n+1)/2 & n \text{ odd,} \end{cases}$$

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from any positive integer eventually reaches 1 [6]. Its negation has two faces: a nontrivial cycle, or a divergent orbit. This paper concerns the divergent face, through the lens of symbolic dynamics. The *itinerary* of n is the infinite binary word

$$w(n) = (T^i(n) \bmod 2)_{i \geq 0},$$

the diary of parity choices the orbit makes. Two facts, both formal in the companion repository [10], frame everything below. First, the itinerary determines the integer completely: distinct positive integers have distinct (in fact everywhere eventually differing) itineraries — the infinite-precision form of Terras’ parity bijection [11]. Second, a divergent orbit cannot have an arbitrary itinerary: its odd-step density is pinned, in the liminf, to at least the critical value

$$\delta_c = \frac{\log 2}{\log 3} = 0.630929\dots$$

(the density floor of [10]; intuitively, fewer odd steps than that and the halvings win).

The ladder. Order itineraries by word-theoretic complexity: eventually periodic words; automatic words (fixed points of constant-length substitutions and their codings); morphic and Sturmian words; and beyond. Excluding each class from the divergent repertoire is a theorem target. The conjecture — on its divergence side — is the top of the ladder; each rung is finite, honest work. This paper establishes the floor and gives the second rung a precise, classical shape.

Results.

1. *Rung zero, machine-verified* (Section 3). A divergent orbit’s itinerary is not eventually periodic. Formalized in Lean 4 (`divergent_itinerary_aperiodic`, file `Ladder.lean` of [10]), depending only on Lean’s standard foundations; no axioms, no `sorry`.
2. *The automatic rung is one Mahler value* (Sections 4–5). For the fixed point w of $\sigma : 1 \mapsto 110, 0 \mapsto 011$, the unique 2-adic integer with itinerary w is $x(w) = -10 + \frac{5}{9}F_2(8/9)$, where $F(z) = \sum_i w_i z^i$ satisfies $F(z) = (z + z^2)/(1 - z^3) + (1 - z^2)F(z^3)$. Hence if $F_2(8/9) \notin \mathbb{Q}$, no integer realizes w . The word w is aperiodic (Lemma 4.1), F is transcendental over $\mathbb{Q}(z)$ (Lemma 4.3), the real value $F(8/9)$ is transcendental by Mahler’s classical theorem (Theorem 7.1), and the statement generalizes uniformly to all constant-length, constant-block-weight substitutions in the supercritical band (Proposition 5.2).
3. *The cheap route is provably insufficient* (Section 6). The truncations of the defining series give rational approximations to $F_2(8/9)$ with irrationality exponent exactly $(3 \log 2)/(2 \log 3) < 1$; rational numbers admit approximations of exponent 1. No argument that uses only the convergence speed of the series can decide irrationality. The deficiency ratio is δ_c again, in disguise, and it caps the naive exponent below 1 throughout the supercritical band — the same coincidence that makes the density theory critical makes the transcendence question non-trivial.

We emphasize the modesty of the target: the automatic rung is a countable, measure-zero family of itineraries, and closing it would leave the conjecture wide open (the companion no-go theorem [10] shows no bounded-memory argument reaches the top of the ladder). Its interest is structural — it is the first rung where the divergence problem meets a developed branch of transcendence theory through an exact interface, with every reduction step proved and most of them machine-checked.

2 Itineraries, the correction term, and realizations

Write T^i for the i -th iterate, $j_T(n)$ for the number of odd values among $n, T(n), \dots, T^{T-1}(n)$ (the odd-step count), and abbreviate the itinerary prefix $w_{<T} = (w_0, \dots, w_{T-1})$, $w_i = T^i(n) \bmod 2$. The dynamics admits an exact linear form (`terras_exact_form` in [10]): for all n, T ,

$$2^T T^T(n) = 3^{j_T} n + d(w_{<T}), \quad (1)$$

where the correction term d depends only on the parity word, via the closed form (`dcoef_closed`)

$$d(w_{<T}) = \sum_{\substack{i < T \\ w_i = 1}} 2^i 3^{\#\{i' > i : i' < T, w_{i'} = 1\}}. \quad (2)$$

Both are theorems of the formal development; we use them as given.

Now let w be an *infinite* binary word. Since 3 is invertible in $\mathbb{Z}_2$, (1) reads $n \equiv -3^{-j_T} d(w_{<T}) \pmod{2^T}$, and from (2),

$$3^{-j_T} d(w_{<T}) = \sum_{\substack{i < T \\ w_i = 1}} 2^i 3^{-(1+\nu_i)}, \quad \nu_i := \#\{i' < i : w_{i'} = 1\},$$

exactly (the exponent in (2) is $j_T - \nu_i - 1$). The right-hand side is a partial sum of a series whose i -th term has 2-adic absolute value 2^{-i} ; it therefore converges in $\mathbb{Z}_2$.

Definition 2.1. The *realization* of an infinite word w is

$$x(w) := - \sum_{i : w_i = 1} 2^i 3^{-(1+\nu_i)} \in \mathbb{Z}_2.$$

Proposition 2.2. A positive integer n has itinerary w if and only if $x(w) = n$ in $\mathbb{Z}_2$. In particular at most one integer realizes any given word.

Proof. If n has itinerary w then $n \equiv -3^{-j_T} d(w_{<T}) \pmod{2^T}$ for every T by (1), and the right-hand sides converge to $x(w)$; hence $n = x(w)$. Conversely if $x(w) = n$ then n agrees with $-3^{-j_T} d(w_{<T})$ modulo 2^T for every T ; by the parity bijection (the first T parities of an orbit are determined by the starting point mod 2^T , `parity_of_modEq` in [10], after [11]) and an induction on T , the itinerary of n is w . $\square$

Excluding a word w from the divergent repertoire therefore means proving $x(w) \notin \mathbb{Z}_{\geq 1}$ — a statement about one 2-adic number per word.

3 Rung zero

Theorem 3.1 (machine-verified). *Let n have unbounded orbit: for every B there is t with $T^t(n) > B$. Then the itinerary of n is not eventually periodic.*

Proof sketch. If the itinerary satisfies $w_{s+t} = w_{s+P+t}$ for all $t \geq 0$, then the two tail points $u = T^s(n)$ and $v = T^{s+P}(n)$ have identical itineraries at every time; by the infinite-precision parity bijection (two integers with everywhere-equal itineraries agree modulo 2^k for every k , hence are equal — `eq_of_itinerary_eq`) we get $u = v$. The orbit is then eventually periodic, so it is bounded by the maximum of its first $s + P + 1$ values, contradicting unboundedness. The full proof is `divergent_itinerary_aperiodic` in `Ladder.lean` of [10]; it depends only on `propext`, `Classical.choice`, `Quot.sound`. $\square$

Rung zero is elementary, but it sets the pattern: a complexity class of words is excluded by showing its realizations are never positive integers (here, eventually periodic words realize the rational 2-adic points of the dynamics — e.g. $x((110)^\infty) = -5$ — and the positive-integer ones among them are the bounded orbits).

4 The automatic rung: one functional equation

From here on, w is the fixed point of

$$\sigma : \quad 1 \mapsto 110, \quad 0 \mapsto 011,$$

starting from $w_0 = 1$: $w = 110110011110110011011110110\cdots$. Both images have length 3 and contain exactly two 1s, so w has parity density exactly $2/3 > \delta_c$: a word in the supercritical band, admissible (as far as the density theory knows) as the itinerary of a divergent orbit. The substitution structure is equivalent to the *kernel relations*

$$w_{3i} = w_i, \quad w_{3i+1} = 1, \quad w_{3i+2} = 1 - w_i \quad (i \geq 0), \quad (3)$$

read off from $\sigma(1) = 110$, $\sigma(0) = 011$.

Lemma 4.1. *w is not eventually periodic.*

Proof. Suppose $w_{i+P} = w_i$ for all $i \geq N$, with $P \geq 1$ minimal admissible. If $3 \mid P$, write $P = 3Q$: for $i \geq N$, (3) gives $w_{i+Q} = w_{3(i+Q)} = w_{3i+P} = w_{3i} = w_i$ for all $i \geq N$, so $Q < P$ is also an eventual period; hence we may assume $3 \nmid P$. Consider positions $i \equiv 1 \pmod{3}$, $i \geq N$; there $w_i = 1$ by (3). If $P \equiv 1 \pmod{3}$ then $i + P \equiv 2 \pmod{3}$, so $1 = w_{i+P} = 1 - w_{(i+P-2)/3}$, i.e. $w_{t+(P-1)/3} = 0$ for every $t \geq (N-1)/3$ (writing $i = 3t + 1$). Then w is eventually all 0s, contradicting $w_{3i+1} = 1$. If $P \equiv 2 \pmod{3}$ then $i + P \equiv 0 \pmod{3}$, so $1 = w_{i+P} = w_{(i+P)/3}$, i.e. $w_{t+(P+1)/3} = 1$ for all large t : w is eventually all 1s, contradicting $w_{3i+2} = 1 - w_i = 0$ at the infinitely many i with $w_i = 1$. $\square$

Proposition 4.2 (the functional equation). *The generating function $F(z) = \sum_{i \geq 0} w_i z^i \in \mathbb{Z}[[z]]$ satisfies*

$$F(z) = \frac{z + z^2}{1 - z^3} + (1 - z^2) F(z^3).$$

Proof. Group the sum in blocks of three and apply (3):

$$F(z) = \sum_{i \geq 0} z^{3i} (w_i + z + (1 - w_i)z^2) = (z + z^2) \sum_{i \geq 0} z^{3i} + (1 - z^2) \sum_{i \geq 0} w_i z^{3i}. \quad \square$$

This is the classical Mahler setup [8, 9]: degree $d = 3$, coefficients in $\mathbb{Q}(z)$, iteration $z \mapsto z^3$. (It is also verified coefficient-wise to degree 900 by the scripts of [10], as are all identities below; see Section 9.)

Lemma 4.3. *F is transcendental over $\mathbb{Q}(z)$.*

Proof. The coefficients of F lie in $\{0, 1\}$. If F were rational, its coefficients would eventually satisfy a fixed linear recurrence with constant coefficients; the sequence of state windows (the D most recent coefficients, for D the recurrence depth) takes values in the finite set $\{0, 1\}^D$, so some window repeats, and forward determinism makes the coefficient sequence eventually periodic — contradicting Lemma 4.1. So F is irrational; by Fatou's theorem [4] (a power series with integer coefficients and radius of convergence 1 is either rational or transcendental over $\mathbb{Q}(z)$; see also [3, Ch. 6]), F is transcendental over $\mathbb{Q}(z)$. $\square$

Theorem 4.4 (the evaluation identity). *In $\mathbb{Q}_2$ (where $|8/9|_2 = 1/8 < 1$ makes all series below converge),*

$$x(w) = -10 + \frac{5}{9} F_2(8/9),$$

where $F_2(8/9) := \sum_{i \geq 0} w_i (8/9)^i \in \mathbb{Z}_2$ denotes the 2-adic evaluation.

Proof. Each block of three letters contains exactly two 1s, so ν_{3t+r} (the number of 1s before position $3t+r$) equals $2t$ plus the count inside the current block. By (3), if $w_t = 1$ the block is $(1, 1, 0)$, with 1s at $r = 0$ ($\nu = 2t$) and $r = 1$ ($\nu = 2t + 1$); if $w_t = 0$ the block is $(0, 1, 1)$, with 1s at $r = 1$ ($\nu = 2t$) and $r = 2$ ($\nu = 2t + 1$). The block contributions to $\sum_{w_i=1} 2^i 3^{-(1+\nu_i)}$ are therefore

$$w_t = 1 : 2^{3t} 3^{-(2t+1)} + 2^{3t+1} 3^{-(2t+2)} = \left(\frac{8}{9}\right)^t \cdot \frac{5}{9}, \quad w_t = 0 : 2^{3t+1} 3^{-(2t+1)} + 2^{3t+2} 3^{-(2t+2)} = \left(\frac{8}{9}\right)^t \cdot \frac{10}{9}.$$

Grouping consecutive terms of a convergent series in a complete ultrametric field is harmless, so

$$x(w) = - \sum_{t \geq 0} \left(\frac{8}{9}\right)^t \left(\frac{10}{9} - \frac{5}{9} w_t\right) = -\frac{10}{9} \cdot \frac{1}{1 - \frac{8}{9}} + \frac{5}{9} \sum_{t \geq 0} w_t \left(\frac{8}{9}\right)^t = -10 + \frac{5}{9} F_2(8/9). \quad \square$$

Remark 4.5. *Iterating Proposition 4.2 at $z = 8/9$ unrolls Theorem 4.4 into a tower: with $u_k := \beta_k^{-1} F_2(\alpha_k/\beta_k)$, $(\alpha_k, \beta_k) = (8^{3^k}, 9^{3^k})$,*

$$u_k = \frac{\alpha_k(\alpha_k + \beta_k)}{\beta_k^3 - \alpha_k^3} + (\beta_k^2 - \alpha_k^2) u_{k+1}, \quad \text{e.g. } u_0 = \frac{136}{217} + 17 u_1,$$

with constants of height $\approx 9^{3^{k+1}}$. The tower view makes the height growth of the method visible; the single-equation view is the one the classical theory consumes.

5 The reduction

Corollary 5.1 (rung 1 for w , reduced). *If $F_2(8/9) \notin \mathbb{Q}$, then no positive integer has itinerary w .*

Proof. If $n \in \mathbb{Z}_{\geq 1}$ had itinerary w , Proposition 2.2 and Theorem 4.4 give $F_2(8/9) = \frac{9}{5}(n+10) \in \mathbb{Q}$. $\square$

The same reduction holds across the natural class.

Proposition 5.2. *Let σ be a substitution on $\{0, 1\}$ of constant length L whose two image blocks each contain exactly a ones, with $2^L < 3^a$ (equivalently: the fixed-point density a/L exceeds δ_c), and let w be a fixed point with generating function F . Then $F(z) = A_\sigma(z) + B_\sigma(z)F(z^L)$ with $A_\sigma, B_\sigma \in \mathbb{Q}(z)$ explicitly computable from the two blocks, and there are rationals A, B (computable from the blocks; $B \neq 0$ unless the two blocks contribute identically, in which case $x(w) \in \mathbb{Q}$ and the question is decidable directly) with*

$$x(w) = A + B \cdot F_2\left(\frac{2^L}{3^a}\right),$$

the series converging both 2-adically ($|2^L/3^a|_2 = 2^{-L}$) and in $\mathbb{R}$ ($0 < 2^L/3^a < 1$, by supercriticality). Hence irrationality of the single 2-adic Mahler value $F_2(2^L/3^a)$ excludes all integers from realizing w .

Proof. Identical to Proposition 4.2 and Theorem 4.4: blockwise grouping. Each block of L letters contains exactly a ones, so $\nu_{Lt+r} = at + (\text{count inside the block})$, the type- b block ($b \in \{0, 1\}$) contributes $z^t C_b$ with $z = 2^L/3^a$ and $C_b = \sum_{r: \sigma(b)_r=1} 2^r 3^{-(1+\#\{r' < r: \sigma(b)_{r'}=1\})}$, and $x(w) = -\sum_t z^t (C_0 + (C_1 - C_0)w_t) = -C_0/(1-z) - (C_1 - C_0)F_2(z)$. For the functional equation, group F in L -blocks as before. The double convergence is the displayed absolute-value computation; $3^a > 2^L$ is exactly $z < 1$ in $\mathbb{R}$. $\square$

The supercritical band — the only band where divergent orbits can live — is thus exactly the band where the Mahler series converges at *both* places. We do not currently use the archimedean convergence, but the classical method does (Section 7), and the coincidence deserves to be on the record.

6 The gap: truncation provably cannot decide

Why is this not finished? The series defining $F_2(8/9)$ converges fast, and fast rational approximation is the standard engine of irrationality proofs. The following proposition shows the engine stalls — by a precisely measurable margin.

Proposition 6.1. *For $N \geq 1$ let P_N/Q_N with $P_N = \sum_{i < N} w_i 8^i 9^{N-1-i} \in \mathbb{Z}$, $Q_N = 9^N$, be the N -term truncation of $F_2(8/9)$. Then:*

1. (quality) $|F_2(8/9) - P_N/Q_N|_2 = 2^{-3i_0(N)}$ where $i_0(N) \in \{N, N+1, N+2\}$ is the first index $\geq N$ with $w_{i_0} = 1$; in particular the error is never 0 and

$$|F_2(8/9) - P_N/Q_N|_2 = Q_N^{-\kappa_N}, \quad \kappa_N \rightarrow \frac{3 \log 2}{2 \log 3} = 0.946394\dots$$

2. (insufficiency) For every rational $\xi = a/b$ and every rational $P/Q \neq \xi$ with Q odd and $|P| \leq Q$,

$$|\xi - P/Q|_2 \geq \frac{1}{(|a| + |b|)Q}.$$

Consequently the approximations of part (1) are consistent with $F_2(8/9) \in \mathbb{Q}$: any family of rational approximations can certify irrationality only with exponent > 1 , and the truncations achieve 0.9464.

Proof. (1) The error is $\sum_{i \geq N} w_i(8/9)^i$; its terms have strictly increasing 2-adic valuations $3i$, so the sum is nonzero with absolute value that of its first nonzero term, 2^{-3i_0} . The word w contains no three consecutive zeros (every block has a 1 in the middle position, by (3)), so $i_0(N) \leq N+2$. Then $\kappa_N = 3i_0(N) \log 2 / (2N \log 3) \rightarrow (3 \log 2) / (2 \log 3)$. (2) $\xi - P/Q = (aQ - bP)/(bQ)$; the numerator is a nonzero integer, so $|aQ - bP|_2 \geq |aQ - bP|^{-1} \geq ((|a| + |b|)Q)^{-1}$, while $|bQ|_2 \leq 1$. $\square$

So the question is genuinely beyond the convergence speed of its own defining series: the gap between 0.9464 and 1 must be crossed by structure, not by speed. Note where the constant comes from. For a general admissible pair (L, a) (Proposition 5.2), the N -term truncation gains $LN \log 2$ bits of 2-adic smallness against a denominator of $aN \log 3$ bits, so the exponent is

$$\kappa = \frac{L \log 2}{a \log 3} = \frac{\delta_c}{a/L},$$

the critical density divided by the word's density — and the supercriticality condition $a/L > \delta_c$ is *exactly* the statement $\kappa < 1$, with equality on the critical line. The constant that creates the divergence problem also protects this corner of it: throughout the band where divergent orbits could live, truncation falls short of irrationality by construction.

7 The route: Mahler's method, and one open question

Mahler's method [8, 9] was invented for precisely the situation of Section 6: values of functions satisfying $F(z) = A(z) + B(z)F(z^d)$ at algebraic points, where naive approximation falls short. Its auxiliary-polynomial construction converts the functional equation into approximation families whose exponent grows with the auxiliary degree — crossing any fixed Liouville threshold — and its hypotheses are, for our instance, verified above: F transcendental over $\mathbb{Q}(z)$ (Lemma 4.3), the point $8/9$ algebraic with orbit $(8/9)^{3^k}$ avoiding the singularities of the coefficients (the poles of $(z + z^2)/(1 - z^3)$ at cube roots of unity and the zeros of $1 - z^2$ at ± 1).

At the archimedean place the classical theorem applies as stated:

Theorem 7.1. *The real number $F(8/9) = \sum_i w_i(8/9)^i \in \mathbb{R}$ is transcendental.*

Proof. Mahler's theorem ([8]; in the form [9, Ch. 1]): if $F \in \mathbb{Q}[[z]]$ converges in $|z| < 1$, satisfies a functional equation of the class above, and is transcendental over $\mathbb{Q}(z)$, then $F(\alpha)$ is transcendental for every algebraic α , $0 < |\alpha| < 1$, whose orbit α^{d^k} avoids the finitely many excluded points (poles and zeros of the coefficient functions). Lemma 4.3 and the singularity check above verify every hypothesis at $\alpha = 8/9$. $\square$

The number rung 1 needs is the *same* function at the *same* rational point, evaluated at the other place where the series converges:

Question 7.2. *Is $F_2(8/9) \in \mathbb{Q}_2$ irrational? More generally: does Mahler's method, transposed to $\mathbb{Q}_2$ — auxiliary polynomials, the non-archimedean Schwarz lemma in place of the maximum principle, heights as before — yield irrationality (or transcendence) of $F_p(\alpha)$ for Mahler functions $F \in \mathbb{Q}[[z]]$ at rational α with $0 < |\alpha|_p < 1$ under the hypotheses of Theorem 7.1?*

We are not aware of a published theorem in exactly this form, though p -adic and function-field variants of Mahler’s method exist in several settings (see [9] and its references; [5] develops the method over function fields in positive characteristic, and [2] raises the structural theory at the archimedean place to a complete description of algebraic relations). Locating the statement, or transposing the archimedean proof — the ingredients are place-generic, and non-archimedean analysis is, if anything, kinder to the relevant lemmas — is the concrete work item this paper isolates. We record what is at stake: a positive answer to Question 7.2 excludes, by Corollary 5.1 and Proposition 5.2 (plus a finite-system extension to shifted fixed points, standard in the Mahler literature [2]), the entire uniform-substitution stratum of the automatic rung.

Two further remarks on the interface. First, the celebrated transcendence results for automatic *numbers* [1] do not apply here: $x(w)$ is not an automatic number — its binary digits are not generated by an automaton; only the *series coefficients* are automatic, and the base mixes powers of 2 against powers of 3. The functional equation is the correct interface, not the digit expansion. Second, the obstruction is not the word combinatorics: all of Sections 2–6 are exact and verified; the single missing ingredient is non-archimedean.

8 Calibration: the $5x + 1$ test

Any principle that would exclude divergence must fail to “prove” the same for the $5x + 1$ map, where orbits (conjecturally, and overwhelmingly in experiment, e.g. from $n = 7$) really do diverge. The present method passes this filter in the correct way. For $5x + 1$, the identical computation realizes the same word w at $x_5(w) = A' + B'F_2(2^L/5^a)$ — the same generating function at a different rational point ($8/25$ for our σ) — and a 2-adic Mahler argument would conclude the same way: *no integer has that exact automatic itinerary under $5x + 1$* . This is consistent with (indeed expected alongside) real divergence: the observed $5x + 1$ escapees have generic-looking itineraries of density near 0.49, nowhere near a substitution fixed point. The method excludes specific symbolic classes; it cannot manufacture global convergence — which is precisely why it can be sound. The $3x + 1$ -specific content remains where the companion papers put it: in the density floor that pins divergent orbits into the narrow band where the ordered words live.

9 Scope, and the ladder above

What would closing rung 1 actually buy? Directly: no divergent orbit has an itinerary that is a (shifted) fixed point of a uniform substitution — a countable family, measure zero in every sense. The conjecture would remain open at every higher rung: morphic words, Sturmian words (where the companion shadow paper locates the slowest-escape candidates), and the unstructured majority, which the no-go theorem of [10] places beyond all bounded-memory arguments. The honest value of the rung is the bridge it builds: an exact, mostly machine-verified chain from the Collatz dynamics to a single value of a classical Mahler function, with the remaining gap isolated as one well-posed question (Question 7.2) in a mature theory — and a quantified certificate (Proposition 6.1) that the question sits genuinely beyond elementary methods, for the problem’s own critical reason.

Reproducibility

The formal results (`eq_of_itinerary_eq`, `divergent_itinerary_aperiodic`, `dcoef_closed`, `terras_exact_form`, the parity bijection) are in `lean/Collatz/Ladder.lean`, `Shadow.lean`, and `Parity.lean` of [10] (Lean 4 + `mathlib`; no axioms beyond the standard three, no `sorry`). The numerical verifications — the functional equation to degree 900, the evaluation identity to 600 bits of 2-adic precision, the exact finite-level tower identities, the block law on 200 random words, the realization digit-tail statistics, and the shadow clustering of automatic words — are `analysis/mahler_experiments.py` (experiments 1–4), deterministic, run time under a minute. The attack plan in extended form, including the failed routes, is `analysis/RUNG1_ATTACK.md`.

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